ECON485 - Homework Assignment 4

NoSQL Concepts and Applications

Task 1: Seat Availability Lookups (Key-Value Database)

In a relational system, checking available seats requires locking tables and performing a `COUNT()` with a mathematical subtraction against the capacity. During peak registration times, thousands of concurrent reads and writes cause massive lock contention and severe lag.

A Key-Value store like **Redis** solves this by keeping the availability counter entirely in RAM. When a student registers, an atomic decrement operation (e.g., `DECR available_seats:section_123`) is executed. Because it is atomic and in-memory, it resolves in less than a millisecond without locking an entire SQL table, completely eliminating the bottleneck while ensuring seats are not overbooked.

Task 3 & 4 Concepts: Historical Action Logs (Document Database)

A Document Store like **MongoDB** is highly effective for storing a student's registration history. Relational tables struggle with history because different actions require entirely different data points. A regular course 'Add' only needs a timestamp and section ID, but a 'Time Conflict Override' requires the conflicting section ID, the approving instructor's name, and an override justification text. In SQL, this leads to tables with many `NULL` columns or overly complex schema designs with multiple sub-tables.

In MongoDB, schema flexibility allows each event to be an independent JSON-like document (BSON) inside a `RegistrationHistory` collection. For example, one document might simply say `{{action: "add", section: 12}}`, while the next says `{{action: "override", section: 14, approved_by: "Dr. Smith", reason: "Graduating senior"}}`.

A student's entire history can be retrieved rapidly via their `StudentID` without expensive `JOIN` operations. Document databases are heavily optimized for append-only workloads, making them ideal for high-throughput action logging where old records are frequently written but rarely updated. Furthermore, nested indexing allows administrators to quickly search through metadata, such as finding all overrides approved by "Dr. Smith", without the rigidity of relational columns.